

POSITION	Research Engineer Real World Inc. (RLWRD)	
CONTACT	5F, L7 Gangnam Tower 415, Teheran-ro, Gangnam-gu, Seoul, Korea E-mail: nahyuk0113@gmail.com	Homepage: https://nahyuklee.github.io/
RESEARCH INTERESTS	3D Computer Vision & Embodied AI , with a focus on: - Visual correspondence learning (<i>i.e.</i>, 2D & 3D feature matching) for point cloud registration and geometric shape assembly - Geometric deep learning (<i>e.g.</i>, SE(3)/SO(3)-equivariant learning) 3D representation alignment and foundation-model-driven 3D perception (<i>e.g.</i>, VLM-based 3D scene understanding) 3D-grounded action models for dexterous robot manipulation	
INDUSTRY EXPERIENCE	Real World Inc. (RLWRD) , Seoul, Korea <i>Research Engineer (Alternative Military Service)</i>	Sep 2026 - present - Supervisor: Jaekyoung Bae and Prof. Jinwoo Shin - Researching robot foundation model (RFM) for dexterous manipulation.
	VUNO Inc. , Seoul, Korea <i>Research Scientist (Alternative Military Service)</i>	Jan 2025 - Sep 2026 - Supervisor: Dr. Sunghoon Joo - Researched geometric deep learning (<i>e.g.</i>, SO(3)-equivariant network) for medical image analysis. - Led key components in VUNO Med[®]-DeepBrain[®], LungCT AI[™], and Fundus AI[™].
RESEARCH EXPERIENCE	Computer Vision and Geometry Group, ETH Zürich , Zürich, Switzerland (Remote) <i>Research Collaborator</i>	Oct 2025 - present - Advisor: Dr. Sunghwan Hong and Prof. Dr. Marc Pollefeys - Researching 3D representation alignment and foundation-model-driven 3D perception for shape assembly and scene understanding [ECCV 2026, CoRL 2026].
	Computer Vision Lab, POSTECH , Pohang, Korea <i>Research Assistant</i>	Jul 2022 - Dec 2024 - Advisor: Prof. Minsu Cho and Prof. Jaesik Park - Researched problems in 3D point cloud matching, with a particular focus on point cloud registration, geometric shape assembly, and equivariant learning [ICCV 2025 Highlight, ICML 2024, IPIU 2024 Best Paper].
EDUCATION	Pohang University of Science and Technology (POSTECH) , Pohang, Korea <i>M.Sc. in Computer Science and Engineering</i>	Feb 2023 - Feb 2025 - Thesis: <i>Combinative Matching for Geometric Assembly</i> - Advisor: Prof. Minsu Cho - Admission with the highest distinction (Bang Sung-Yang Fellowship)
	Chung-Ang University , Seoul, Korea <i>B.Sc. in Computer Science and Engineering</i>	Mar 2019 - Feb 2023 - Advisor: Prof. Junseok Kwon - Cumulative GPA: 4.31/4.5 (Summa Cum Laude, 1st in Class of 2023)
	Hyundai Chung-Un High School , Ulsan, Korea <i>Natural Science Track</i>	Mar 2015 - Feb 2018

PUBLICATIONS	All papers are available on https://nahyuklee.github.io/publications/. * denotes equal contribution ; † denotes equal advising. [C4] Zhiang Chen*, Nahyuk Lee*, Boyang Sun, Taein Kwon, Marc Pollefeys, Zuria Bauer†, Sunghwan Hong†, “PROSE: Training-Free Egocentric Scene Registration with Vision-Language Models,” in <i>Proceedings of the Conference on Robot Learning (CoRL)</i>, 2026 [C3] Nahyuk Lee*, Zhiang Chen*, Marc Pollefeys, Sunghwan Hong, “TORA: Topological Representation Alignment for 3D Shape Assembly,” in <i>Proceedings of the European Conference on Computer Vision (ECCV)</i>, 2026 [C2] Nahyuk Lee*, Juhong Min*, Junhong Lee, Chunghyun Park, Minsu Cho, “Combinative Matching for Geometric Shape Assembly,” in <i>Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)</i>, 2025, Highlight (263/11,152 $\approx$ 2.35%) [C1] Nahyuk Lee*, Juhong Min*, Junha Lee, Seungwook Kim, Kanghee Lee, Jaesik Park, Minsu Cho, “3D Geometric Shape Assembly via Efficient Point Cloud Matching,” in <i>Proceedings of the International Conference on Machine Learning (ICML)</i>, 2024 [W1] Seung Hyun Lee, Nahyuk Lee, Chanyoung Kim, Wonjeong Ryoo, Jinkyu Kim, Sang Ho Yoon, Sangpil Kim, “Audio-Guided Image Manipulation for Artistic Paintings,” in <i>NeurIPS Workshop on Machine Learning for Creativity and Design (NeurIPS Workshop)</i>, 2021
PROFESSIONAL SERVICES	Reviewer of international journals IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) Reviewer of international conferences Neural Information Processing Systems (NeurIPS) AAAI Conference on Artificial Intelligence (AAAI) International Conference on 3D Vision (3DV) LSRW Workshop, Conference on Robot Learning (CoRLW) WiCV Workshop, Asian Conference on Computer Vision (ACCVW)
HONORS AND AWARDS	Travel Grant, IEEE/CVF International Conference on Computer Vision (ICCV), 2025 - Waived registration fees with partial scholarship from ICCV 2025, \$1,100 (total) Best Paper Award (Grand Prize, 1st of 311 papers $\approx$ 0.3%), 36th Workshop on Image Processing and Image Understanding (IPIU), 2024 18th Kwanjeong Educational Foundation Fellowship, KEF, 2023 - Winner, received \$17,000 during 2 years of my graduate studies Bang Sung-Yang Graduate Fellowship, POSTECH, 2023 - Admission with highest distinction, \$1,200 16th Kwanjeong Educational Foundation Fellowship, KEF, 2021 - Winner, received \$16,500 during 2 years of my undergraduate studies Da Vinci Scholarship, CAU, 2019 - Admission with highest distinction, \$7,250 Dean’s List, CAU, 2019 - 2021 - 2019 Fall, 2020 Spring, 2020 Fall, 2021 Fall (4 times)
TEACHING EXPERIENCE	Pohang University of Science and Technology (POSTECH) <i>Teaching Assistant</i> Feb 2024 - Jun 2024 • CS537/AIGS537 - Artificial Intelligence & Data Science (2024 Spring)

Ministry of National Defense, Republic of Korea

Instructor

Nov 2022 - Dec 2022

- Conducted lectures on AI projects for South Korean soldiers and short-term service officers
- Collaborated with *Elice, Inc.*

Samsung Electronics Co., Ltd

Instructor

Aug 2022 - Oct 2022

- Conducted lectures on data science and machine learning for newcomers (DS division)
- Collaborated with *Elice, Inc.*

LEADERSHIP

Graduate School of Dept. of CSE, POSTECH

President of Student Council

Feb 2024 - Feb 2025

- President of graduate school council of Department of Computer Science & Engineering, POSTECH.

Kwanjeong Educational Foundation Fellowship

Vice President

Jul 2023 - Jul 2024

- Vice president of 18th Kwanjeong Educational Foundation Fellowship mensches.

TALKS

Shaping Visual Intelligence: The Past, Present, and Future of Computer Vision, Kangwon National University, Samcheok, Korea, Nov. 2025

LLMs as Research Assistants: A Practical Guide from My Latest Papers, VUNO R&D Seminar, Seoul, Korea, Jun. 2025

3D Geometric Shape Assembly via Efficient Point Cloud Matching, Samsung AI Forum (**SAIF**) 2024, Suwon, Korea, Nov. 2024

Combinative Matching for Shape Assembly, Samsung Global Technology Symposium (**GTS**) 2023, Seoul, Korea, Mar. 2023

LANGUAGE

Korean (native), English (fluent)

SKILLS

REFEREES

Available upon request.